

Mandatory Requirements for the Well Drilling Rig

I. Core System Capabilities & Operational Requirements

These high-level requirements are critical for deployment and operation in remote island environments.

- **Complete Drilling System:** The procurement is for a complete, fully operational Water Well Drilling Machine System. This includes the Prime Mover Tractor Rig, Drill Trailer, Mud Recycling System, Rig Tender Truck, Drill Site Tender Trailer, and all other specified components or drilling system with same or better capabilities.
- **Drilling Performance:**
 - Capable of both air and mud rotary drilling to a total depth (TD) of 1,500 feet.
 - Minimum borehole diameter of 12 ¼ inches to total depth.
 - Hoisting capability of at least 50,000 lbs.
- **Air Transportability:**
 - The entire system must be air transportable via C-17 or C-5 aircraft.
 - Must support drive-on/drive-off loading without the use of external material handling equipment or approach shoring.
 - Major components (e.g., mast, mud pump) shall not require removal for air transport. Compliance with MIL-STD-1791 and MIL-STD-209 is required.
- **Engine Type:** All diesel engines must be "Export Engines" (Tier 3 Diesel or equivalent) capable of operating worldwide without Ultra-Low Sulfur Diesel (ULSD). This is a mandatory requirement with no exceptions.
- **Corrosion Protection:** The system must feature the manufacturer's maximum corrosion/rustproof protection package, with a minimum 5-year life expectancy. All components must be painted Desert Sand (No. 30313 FED-STD-595).
- **Self-Sufficiency:** Capable of continuous operations for a minimum of 30 hours without servicing, other than refueling. The system must include integrated support equipment like a welder, toolboxes, and fluid transport (water/fuel).
- **Spare Parts:** Include any expected replacement parts for a repair kit that are likely to fail over time.

II. Salient Characteristics (by Component)

The following are mandatory minimum specifications for the major components of the system.

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Component	Characteristic	Requirement
Prime Mover	Engine	275-350 HP, Tier 3 Diesel "Export Engine".
	Drivetrain	All-Wheel Drive (4x4) with limited-slip or locking differentials.
	Fuel Capacity	200-gallon minimum.
	Cold Weather Ops	Fully operational down to -25°F.
Drill Rig Engine	Engine	375 HP, Tier 3 Diesel "Export Engine".
Top Head Drive	Torque	70,000 in-lb minimum.
	Rotation Speed	0-120 RPM, variable.
Mast	Capacity	100,000 lbs minimum rated working capacity.
Feed System	Pullback Force	50,000 lbs minimum.
	Down Pressure	25,000 lbs minimum.
Main Hoist	Capacity	50,000 lbs minimum via a two-part line.
Mud Recycling System	Capacity	500 GPM processing, 2,500-gallon tank.
	Power	Self-contained, powered by its own Tier 3 Final diesel generator.
Auxiliary Air Compressor	Rating	900 CFM @ 500 PSI & 1150 CFM @ 350 PSI.
	Engine	Tier 3 Diesel Engine.

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Rig Tender Truck	Water Tank	2,000-gallon capacity.
	Auxiliary Fuel	230-gallon capacity with transfer pump and reel.
	Crane	14,000 lbs minimum capacity at a 3-foot radius.
	Welder	250-amp diesel welder/generator with full support kit.
	Winch	20,000 lb front-mounted winch.
	Towing	33-ton (or greater) capacity pintle hook with full trailer connections.
Drill Site Tender Trailer	GVWR	30,000 lbs.
	Deck	20 ft x 8 ft flatbed with tilt and winch capabilities.
Grout Machine	Performance	0-30 GPM at pressures up to 500 PSI.
	Hose Reel	Must hold 1,000 ft of 1 ¼-inch grout hose.
Duplex Mud Pump	Performance	300 GPM at 400 PSI.
	Mounting	Must be rig-mounted.

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III. Mandatory Services & Support

The following services and deliverables are required from the vendor at the time of delivery.

- Training:
 - Minimum 40 hours of on-site familiarization training for operators.
 - Minimum 40 hours of on-site familiarization training for maintenance personnel.
 - Note: All transportation are to be in accordance with the Joint Federal Travel Regulation (JFTR). For example, air travel shall ONLY be allowable using economy class rates that can be found from various on-line travel agents.
- Manuals & Data:
 - One complete hard copy set and one electronic (IPDF) set of all operator, parts, and service manuals for every component and engine.
 - Delivery of manuals is required at the same time as vehicle delivery.
- Diagnostics:
 - Vendor must provide all diagnostic software and devices required to allow Air Force personnel to locally troubleshoot and service all system components.
 - Any software provided must be compatible with Windows 10 (64-bit) and be free of vulnerabilities.
- Warranty:
 - A comprehensive warranty covering all components, regardless of manufacturer, for 24 months or 1,000 operating hours, whichever comes first (or the maximum warranty available if greater).
- Marking:
 - Application of standard military data plates with required information (Nomenclature, S/N, Weight, etc.).
- Export Lift Kits:
 - Lifting kits must be provided for both the rig tender truck and the drill site tender trailer to facilitate overseas shipment.